SolarChain-Eval: A Physics-Constrained Benchmark for Trustworthy Economic Agents in Decentralized Energy Markets

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Abstract

Scalar reward is insufficient for evaluating autonomous economic agents whose decisions affect physical resources. We introduce SolarChain-Eval, a benchmark for decentralized peer-to-peer solar markets that couples reinforcement-learning policies to a physics oracle. An evaluation-only LLM Planner/Auditor provides bounded oversight of proposed actions. The protocol relates episode utility to physical invalidity and its market consequences. Separate diagnostics characterize execution friction and temporal actions; city-level allocation and oversight traces make distribution and intervention visible. Across three archived seeds, 90 rollouts per policy show a utility–safety trade-off: removing the physics penalty increases apparent return while increasing artificial liquidity for learned policies. Planner/Auditor interventions make selected corrections traceable, but do not compensate for reward misspecification. The accompanying artifact package preserves the archived inputs, checkpoints, logs, source snapshot, and scripts needed to reproduce the reported summaries.

1 Introduction

Autonomous agents are increasingly used where digital decisions affect physical resources and market outcomes [Xi et al., 2025, Vyas and Mercangöz, 2025]. Peer-to-peer solar markets make the resulting evaluation problem concrete. An agent may increase apparent utility by allocating rewards or liquidity to generation that exceeds photovoltaic capacity or fails verification. The simulator may then prefer an economically attractive policy even though part of its market activity is not physically deliverable. Local weather and panel capacity determine feasible generation, while verification and settlement determine how that generation enters the market [Zhang et al., 2018, Mengelkamp et al., 2018].

Evaluation must therefore follow the full decision loop rather than stop at episode return. Physical accounting reveals whether the policy backs invalid supply; market and temporal diagnostics show what that decision does to liquidity, execution, and action behavior. When an oversight layer changes an action, the intervention must also leave enough evidence to reconstruct what changed and why. General risk-management and safe-control guidance motivates this broader view, but the relevant evidence must be defined for the energy-market mechanism itself [National Institute of Standards and Technology, 2023, Drgona et al., 2025, Phiri, 2025].

We present SolarChain-Eval, a reproducible benchmark that couples a Gymnasium-compatible economic environment with a physics oracle and an evaluation-only Planner/Auditor interface. The benchmark uses the same data and environment across static, heuristic, PPO, SAC, and DQN policies. A no-physics-penalty ablation exposes reward misspecification by removing the physics term from optimization while continuing to measure violations. A bounded Planner/Auditor layer then tests whether deployment-time oversight can revise risky actions and produce structured evidence without changing the trained policy.

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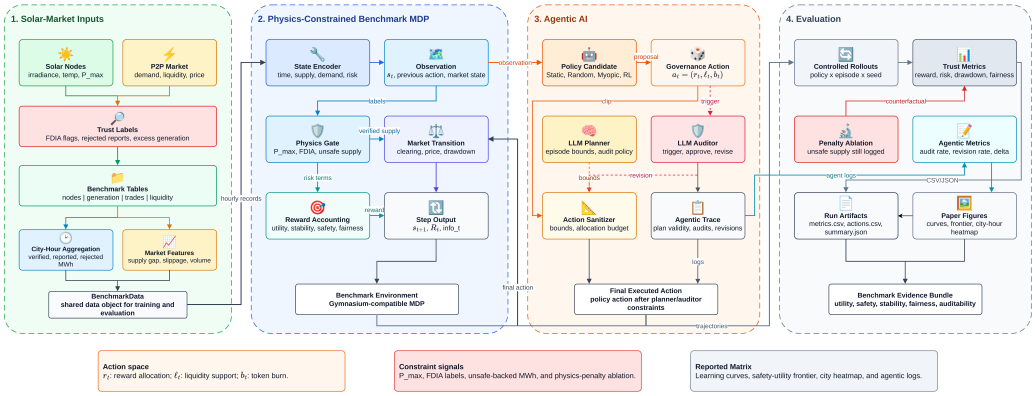


Figure 1: Overall SolarChain-Eval benchmark pipeline. The figure summarizes the end-to-end workflow from data loading, RL training, policy evaluation, output generation, and trustworthiness evidence for decentralized energy-market governance.

The benchmark addresses three questions. **RQ1** compares the utility of learned market policies with their physical and market behavior relative to static and heuristic controls. It also examines temporal action behavior and city-level allocation balance. **RQ2** asks whether reward-maximizing policies back invalid generation and create artificial liquidity when the physics penalty is removed. **RQ3** asks when a bounded LLM Planner/Auditor changes risky actions, what evidence it records, and which failures remain because the underlying reward is misspecified.

SolarChain-Eval makes invalid generation economically observable by measuring the artificial liquidity created when a policy backs suspicious supply. Its evaluation protocol covers three archived seeds, with 90 rollouts per policy and paired comparisons between constrained and no-physics settings. The benchmark also treats LLM oversight as a bounded evaluation interface, keeping it separate from the constraints used to train the policy.

The results are exploratory benchmark evidence. Learned policies improve episode return relative to random and heuristic controls, but no algorithm is uniformly preferable once physical backing and market friction are considered. Removing the physics penalty raises both return and artificial liquidity. Planner/Auditor intervention is selective, traceable, and policy-dependent. Taken together, the results separate two roles: training constraints limit what the optimizer can exploit, whereas structured oversight bounds and records selected deployment-time corrections.

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2 Related Work and Benchmark Positioning

Reinforcement learning for market governance. Reinforcement learning (RL) provides a natural formulation for sequential incentive allocation and liquidity control. PPO constrains on-policy updates through a clipped objective [Schulman et al., 2017]. SAC supports continuous exploration through entropy regularization [Haarnoja et al., 2018], whereas DQN supplies a value-based baseline for discretized actions [Mnih et al., 2015, Van Hasselt et al., 2016]. Stable-Baselines3 provides a common implementation interface for these algorithms [Raffin et al., 2021]. Energy applications include peer-to-peer exchange, distribution-system control, and demand response [Chen et al., 2022, Kelly et al., 2020, Lozano and Shi, 2025]. In decentralized markets, however, policy performance also depends on how settlement rules and liquidity interact with prosumer incentives [Tushar et al., 2018, Mengelkamp et al., 2018].

In energy markets, the action objective is coupled to localized generation, demand, and network or verification constraints. Reward-centric comparisons can miss violations hidden by economic aggregation, while unreliable implementation and limited seed reporting can make RL conclusions difficult to interpret [Henderson et al., 2018, Dulac-Arnold et al., 2021]. Constrained RL addresses feasibility during optimization [Achiam et al., 2017], and safe-control research emphasizes explicit

Positioning summary: representative RL–LLM governance patterns and the SolarChain-Eval execution boundary. The comparison describes roles and limitations; it does not rank prior systems.

Framework	Domain	Upper-level role	Boundary or limitation
RE-GoT [Yao et al., 2025]	Robotic control	Evolves reward functions and interprets rollouts	Reward revision can be expensive in a real-time loop
RL^2 [Yang et al., 2025b]	Distribution networks	Translates operator safety intent into DRL penalties	Numerical feasibility still depends on the lower-level controller
ADMM–LLM [Yang et al., 2025a]	Microgrid coordination	Adjusts decentralized optimization parameters	Heuristic updates do not replace convergence or constraint checks
SolarChain-Eval	P2P solar market	Bounds and audits fixed RL actions during evaluation	Physics oracle remains authoritative; logs support paired diagnosis

physical models and verifiable operating limits [Drgona et al., 2025]. SolarChain-Eval complements these directions with an evaluation benchmark: it records how much invalid supply a policy backs, how that supply enters liquidity, and how utility changes when the physics term is removed.

RL–LLM integration and evaluation. Recent bi-level systems assign language models a slower supervisory role over learned or numerical controllers [Yao et al., 2022, Du et al., 2023, Carta et al., 2023]. Language feedback has been used to evolve rewards [Yao et al., 2025, Ma et al., 2023], translate operator intent into lower-level penalties [Yang et al., 2025b], and adjust decentralized optimization parameters [Yang et al., 2025a]. Such interfaces can expose risk signals and intervention rationales to inspection. They do not, by themselves, guarantee numerical feasibility: a plausible explanation may still accompany an invalid action unless execution remains subject to deterministic checks [Rudin, 2019, Amodei et al., 2016].

SolarChain-Eval isolates this question by using the LLM only at evaluation time. The Planner emits bounded action ranges and audit rules; the Auditor is called under event triggers and returns a schema-validated approval or revision. The environment remains the final authority on action sanitization and physical accounting. This differs from systems that use an LLM to rewrite the training objective: the archived RL checkpoints remain fixed, which makes the comparison about bounded deployment-time processing rather than retraining.

These strands leave a narrower evaluation gap: existing tools do not necessarily reveal when economically rewarded behavior depends on invalid physical supply, or how a deployment-time correction relates to that failure [Liu et al., 2023, Li et al., 2023, Phiri, 2025]. SolarChain-Eval addresses this gap as a benchmark rather than a new RL algorithm or a general governance guarantee. It links economic outcomes to physical accounting and preserves the traces needed to inspect bounded interventions from fixed artifacts.

3 SolarChain-Eval Benchmark

3.1 Market setting and physics-constrained MDP

Each episode is a 24-hour market cycle sampled from an April 2026 five-city dataset. The observation combines time-of-day features with verified and reported generation and the physical PV upper bound. Market state is represented by the supply–demand gap, liquidity, and token price; risk and action history are carried by the remaining channels. The action is

$$s_t = [\sin\tau_t, \cos\tau_t, G_t^v, G_t^r, P_t^{\max}, \Delta_t, L_t, p_t, \nu_t, \sigma_t, \alpha_{t-1}, \ell_{t-1}], \quad a_t = (\alpha_t, \ell_t, b_t), \tag{1}$$

where α_t allocates backed supply to rewards, ℓ_t allocates it to liquidity, and b_t is the token burn rate. The sanitizer enforces $\alpha_t + \ell_t \leq 0.98$ and the configured component bounds. PPO and SAC use the continuous action representation; DQN selects one of $5^3 = 125$ grid points.

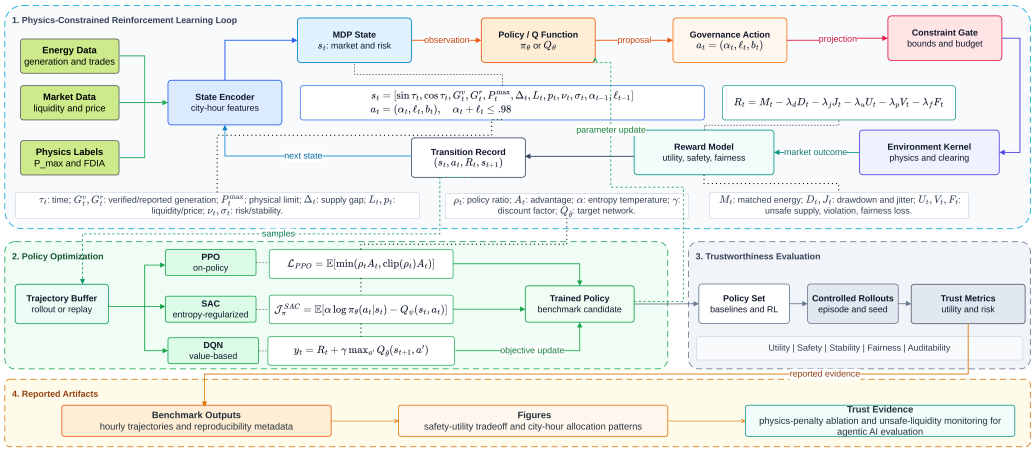


Figure 2: Reinforcement learning evaluation framework for SolarChain-Eval. The diagram connects market and physical inputs to policy actions, environment feedback, PPO/SAC/DQN updates, and trustworthiness-oriented evaluation across utility, safety, stability, smoothness, and fairness.

The state is assembled at the hourly market boundary, so all policies observe the same information and episode start rules. G^v_t is verified generation and G^r_t is reported generation. $P^{\max}_t$ is the PV capacity implied by irradiance and the installed panel specification after temperature adjustment. The gap and risk channels expose the information available to a market operator, while the previous action ratios make action smoothness measurable rather than implicit. Static, Random, and Myopic controls use the same observation and sanitizer but do not update parameters.

The physics module marks reported generation that exceeds the PV upper bound or carries an FDIA label. Let X_t be suspicious supply and let $\beta_t = \min(\alpha_t + \ell_t, 0.98)/0.98$. The action-dependent violation rate is

$$V_t = \frac{\beta_t X_t}{\max(G^v_t + \beta_t X_t, \epsilon)}, \qquad A_t = \beta_t X_t. \tag{2}$$

The market clears effective demand after token burning, updates residual liquidity, and records artificial liquidity as $\ell_t A_t$. Thus the benchmark distinguishes a physical anomaly from the amount of anomalous supply that a policy economically backs.

For clarity, the benchmark treats a violation and artificial liquidity as related but distinct quantities. A generation record can be invalid before any policy acts. It becomes artificial liquidity only when the action allocates part of the market budget to that suspicious supply. This distinction prevents a policy from appearing safe merely because the environment reports a low anomaly count, and it allows the ablation to test whether reward optimization changes how aggressively invalid supply is backed.

3.2 Reward and evaluation metrics

The step reward combines market clearing and penalties:

$$R_t = M_t - \lambda_d D_t - \lambda_j J_t - \lambda_u U_t - \lambda_p (V_t + A_t) - \lambda_f F_t. \tag{3}$$

Here M_t is matched energy. The penalties cover liquidity drawdown D_t, action jitter J_t, unmet demand U_t, the physics terms $V_t + A_t$, and city-level reward imbalance F_t. The no-physics setting sets $\lambda_p = 0$ while continuing to log V_t and artificial liquidity. At the episode level, the benchmark reports return and matched volume together with violation rate and artificial liquidity. Slippage and drawdown describe market friction, action jitter captures temporal variation, and spatial fairness summarizes city-level allocation balance.

The reported fairness statistic is computed from city-level reward allocations over an episode; it is a benchmark balance diagnostic, not a population-level equity estimate. Slippage is retained as a market-friction measure, while drawdown captures whether the policy consumes liquidity faster than the environment replenishes it. We report all metrics even when they are not part of the optimization objective, because a scalar reward can conceal a trade-off among these dimensions.

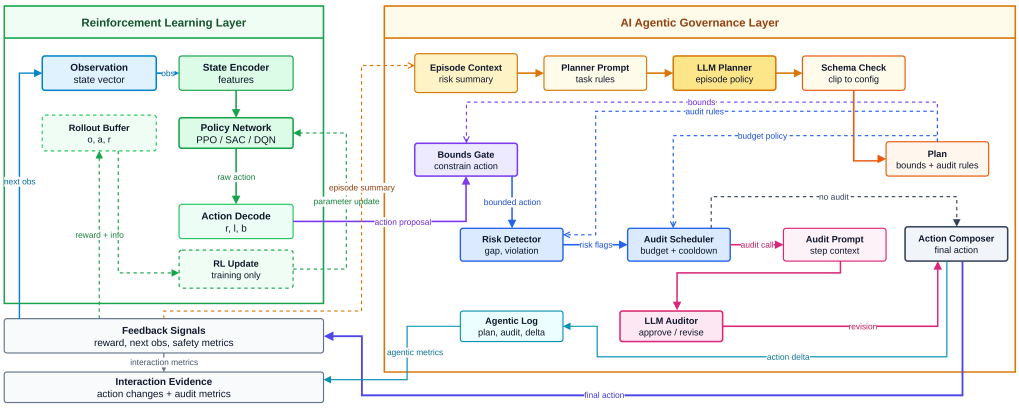


Figure 3: Interaction between the reinforcement-learning layer and the evaluation-only Planner/Auditor layer. The policy proposes market actions, which are bounded by the Planner and conditionally revised by the Auditor under high-risk scenarios.

3.3 Evaluation-only Planner/Auditor

The agentic layer is inserted between a trained policy and the environment only during evaluation. At episode start, the Planner proposes component bounds and an audit policy defined by its budget, trigger thresholds, and cooldown. The proposed action is clipped to these bounds at every step. High raw violation, supply shortfall, and high static slippage are hard triggers; excessive action jitter is a soft trigger. When an eligible event occurs, the Auditor returns either `approve` or a schema-validated `revise` action. The wrapper records the trigger context, both action versions, the resulting delta, and the Auditor decision with its rationale. Invalid or unparsable LLM outputs fail the formal run rather than silently controlling the market.

This separation defines the scope of the agentic claim. Figure 3 summarizes this execution boundary. The Planner/Auditor does not modify the policy network, reward weights, market data, or physics oracle. It can bound or revise a proposed action after the policy has been trained, and it can produce an inspectable trace of why a revision occurred. Hard action constraints remain in the environment, so an accepted natural-language rationale never directly controls market execution. The layer is therefore evaluated as a bounded governance interface, not as a substitute for constrained optimization or human regulatory review.

4 Experimental Protocol

4.1 Data, policies, and archived seeds

The benchmark covers Beijing, Shanghai, Chengdu, Shenzhen, and Hangzhou from 2026-04-01 through 2026-04-30. Its 50 energy nodes yield 720 hourly market states and 36,000 generation records. The archive also contains 1,185 P2P trades and 1,800 FDIA-marked generation rows. Each policy is trained for 300,000 timesteps and evaluated for 30 episodes per seed. Seeds 20260511, 20260512, and 20260513 therefore provide 90 rollouts per policy. All three runs share the same dataset digest and configuration; the first predates a progress-display-only code change documented in the artifact manifest.

We compare Static, Random, Myopic, PPO, SAC, and DQN in the main setting and no-physics ablation. The agentic setting evaluates PPO, SAC, and DQN with an evaluation-only LLM Planner/Auditor. Agentic main and no-physics runs each contain 90 rollouts per learned policy and 2,160 governance log rows per policy. All tables report mean $\pm$ standard deviation across rollout-level records.

The three archives define the evidence boundary for this revision. Seed 20260511 predates a progress-display-only code change, but its metadata preserve the same scientific configuration and data digest as the two later archives. We do not pool intermediate runs or add significance tests that are absent from the archived analysis. Nor do we claim that the entire pipeline was independently rerun.

Table 1: Benchmark data, training, and evaluation configuration used by the archived runs. The expanded view makes explicit which quantities describe the market instance, which describe policy training, and which determine the reported evidence.

Category	Quantity	Setting	Evidence role
Dataset	5 cities; 50 nodes	Beijing, Shanghai, Chengdu, Shenzhen, Hangzhou	Spatial coverage and node heterogeneity
Temporal data	720 hourly states	2026-04-01–2026-04-30	30 daily episodes per seed
Physical records	36,000 generation; 1,800 FDIA rows	PV upper bounds and verification labels	Physics oracle and violation accounting
Market records	1,185 P2P trades; 720 liquidity states	Demand, liquidity, price, slippage	Market clearing and friction metrics
Policy training	300,000 timesteps per seed	PPO, SAC, DQN	Archived learned controllers
Baselines	Static, Random, Myopic	Same environment and action sanitizer	Non-learning comparison controls
Seeds	20260511, 20260512, 20260513	Same dataset digest and configuration	Cross-seed variability
Evaluation	30 episodes per seed; 90 rollouts per policy	Mean $\pm$ standard deviation; paired episode keys	Main, ablation, and agentic comparisons
Governance logs	2,160 steps per policy and setting	Event-triggered Planner/Auditor	Auditability and intervention evidence

Paired comparisons match seed, policy, and episode before computing no-physics-minus-physics or RL+LLM-minus-RL deltas.

4.2 Evaluation settings and provenance

The main benchmark uses the full physics-constrained reward, whereas the ablation removes only its physics penalty. The agentic setting reuses the trained policies and applies bounded Planner/Auditor processing during evaluation; it does not retrain them. The artifact package preserves all three seed archives together with the metrics, model checkpoints, governance logs, configurations, and source snapshot. It also contains the scripts used for aggregation and figure generation. No additional experiments are claimed in this revision.

For the main and ablation settings, 540 rollout rows are present per setting across all six policies. Each learned-policy agentic setting contains 270 rollout rows and 2,160 hourly governance records per policy. The scripts preserve seed labels, setting names, policy names, and episode identifiers in the derived tables. This makes the reported standard deviations and confidence intervals traceable to explicit rollout-level units rather than to visually aggregated pixels.

5 Results

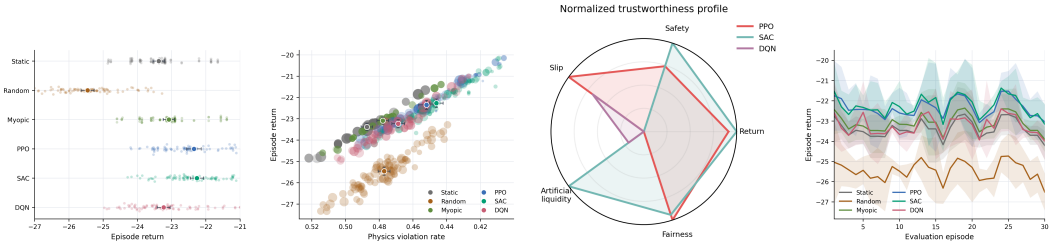
5.1 Main benchmark performance

Table 2 summarizes 90 rollouts per policy under the physics-constrained reward. PPO and SAC achieve the strongest episode returns among learned policies. SAC also has the highest matched volume and the lowest artificial liquidity in this aggregate. These gains do not identify a uniformly preferable policy: SAC's lower artificial liquidity is accompanied by higher mean slippage, whereas PPO has lower slippage but more artificial liquidity. DQN improves on Random for return and volume but remains close to the heuristic controls on violation and friction. The comparisons are descriptive and do not imply that one algorithm dominates across the full evaluation.

The figures explain why the table should not be reduced to a return ranking. Figure 4 relates rollout-level rewards to the utility–safety frontier and their episode trajectory. Figure 5 then separates physical violation from the artificial liquidity and slippage observed in the same runs. A policy can move

Table 2: Main benchmark under the physics-constrained reward. Values are mean ± standard deviation over 90 rollouts per policy across three archived seeds.

Policy	Episode return	Volume	Violation rate	Slippage	Artificial liquidity (MWh)
Static	-23.38 ± 0.80	0.514 ± 0.099	0.4874 ± 0.0166	0.0140 ± 0.0020	0.1803 ± 0.0419
Random	-25.46 ± 0.91	0.480 ± 0.093	0.4771 ± 0.0172	0.0148 ± 0.0023	0.1137 ± 0.0297
Myopic	-23.08 ± 0.80	0.456 ± 0.088	0.4779 ± 0.0168	0.0144 ± 0.0023	0.1009 ± 0.0235
PPO	-22.35 ± 1.02	0.470 ± 0.097	0.4520 ± 0.0219	0.0134 ± 0.0023	0.1453 ± 0.0840
SAC	-22.26 ± 0.94	0.521 ± 0.100	0.4461 ± 0.0196	0.0178 ± 0.0032	0.0686 ± 0.0328
DQN	-23.23 ± 0.91	0.491 ± 0.096	0.4688 ± 0.0182	0.0148 ± 0.0023	0.1298 ± 0.0397



(a) Reward distribution. (b) Utility–safety frontier. (c) Normalized profile. (d) Reward trajectory.

Figure 4: Main benchmark performance under the physics-constrained reward. Panels (a), (b), and (d) are the original rollout-level figures; panel (c) is the only newly drawn summary panel and uses min–max normalization across learned policies.

toward higher utility while shifting risk to another part of the market mechanism, which reward-only reporting would conceal.

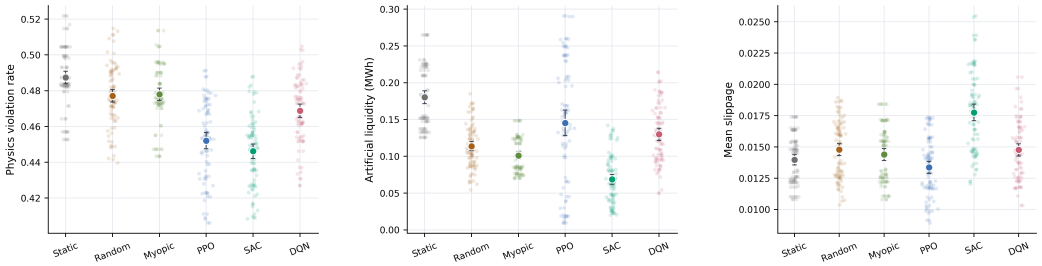
5.2 Physics-penalty ablation

Removing the physics penalty increases scalar episode return for all learned policies, but the same rollouts also show higher artificial liquidity and violation rates. The paired deltas are $+0.0785 \pm 0.0678$ MWh for PPO, $+0.1238 \pm 0.0386$ MWh for SAC, and $+0.0576 \pm 0.0317$ MWh for DQN. With the LLM layer active, the corresponding artificial-liquidity deltas are $+0.0598 \pm 0.0631$, $+0.1149 \pm 0.0388$, and $+0.0413 \pm 0.0317$ MWh. Matching by seed, policy, and episode isolates the within-benchmark change associated with the reward specification. It does not establish a causal effect beyond these archived comparisons.

The direction of this comparison is more informative than the absolute reward scale. In the constrained setting, artificial liquidity is lower for the learned policies than in the no-physics setting even though the latter produces a higher scalar return. The original ablation panels in Figure 6 show this effect both as paired deltas and as a trajectory across the 24-hour episode. The result is consistent with reward misspecification: removing a penalty changes what the optimizer treats as profitable, while logging the violation metric after training reveals the hidden cost.

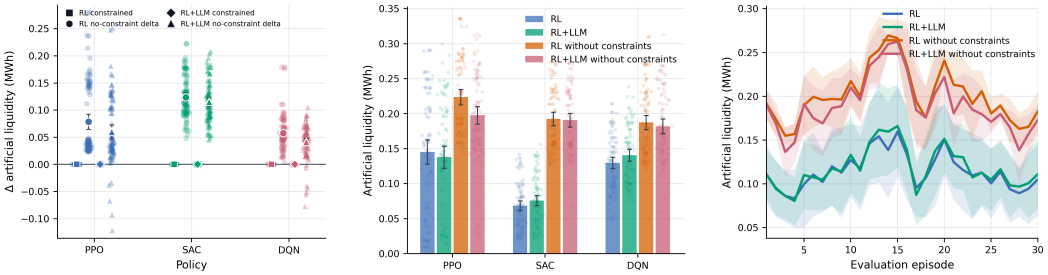
5.3 Agentic governance diagnostics

Under the constrained reward, Planner validity is 1.0 for all learned policies. Audit calls occur on roughly 34–37% of hourly steps, and more than 94% of those calls return revisions in the aggregate. Their effects depend on the base policy. Paired RL+LLM-minus-RL action-jitter deltas are positive for PPO and negative for SAC and DQN, while changes in artificial liquidity remain small relative to the no-physics ablation. Figure 7 presents the audit rates beside paired episode deltas and step-level Auditor changes. Its compact panels allow the two action-delta views to be compared without distortion. The evidence supports a narrow conclusion: structured oversight makes selected interventions visible, but does not uniformly improve the underlying objective.



(a) Physics violation rate. (b) Artificial liquidity. (c) Mean slippage.

Figure 5: Trustworthiness metrics in the main benchmark. The panels are the original result images and share a fixed subfigure height for direct visual comparison; translucent points denote rollout-level observations.



(a) Paired reward and risk deltas. (b) Artificial-liquidity levels. (c) Episode trajectory.

Figure 6: Physics-penalty ablation. The original paired-rollout, setting-comparison, and trajectory images are shown with equal subfigure heights. Positive deltas indicate that removing the physics penalty increases the corresponding quantity.

The logs also clarify what this selectivity means. The Planner is valid for every learned-policy rollout, but the Auditor is called on only a subset of hourly decisions. A revision rate above 94% under the constrained setting means that eligible calls usually change the proposed action; it does not mean that most actions are revised. PPO receives more frequent and larger corrections, whereas SAC and DQN show reductions in action jitter. Audit frequency must therefore be read together with whether a call revises the action and by how much. Collapsing these records into one governance score would obscure the policy dependence visible in the paired panel.

5.4 Representative audit trace

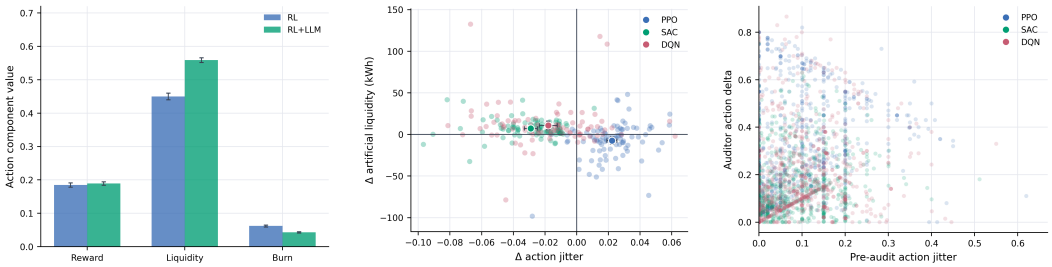
The aggregate governance statistics are complemented by a trace-level view. In the archived no-physics PPO trace, the environment reports a raw violation rate of 0.14 and a supply gap of 1.97 when the policy proposes $(r = 0.049, \ell = 0.931, b = 0.200)$. The Auditor revises the action to $(r = 0.050, \ell = 0.200, b = 0.000)$, producing an action delta of 0.932. This case is not a causal intervention study; it is a reproducible example of how the wrapper records the trigger context, proposed action, validated final action, and rationale.

6 Discussion, Limitations, and Implications

The results separate two safeguards that act at different points in the decision loop. Physics constraints operate during optimization and prevent invalid supply from being rewarded as ordinary market inventory. Planner/Auditor oversight acts only after the policy proposes an action, where it provides a bounded and inspectable correction channel. Removing the physics penalty raises apparent utility and artificial liquidity; subsequent oversight can revise selected actions but cannot repair the objective that produced them. A post-hoc auditor can therefore document and bound an action, but it cannot make a misspecified training objective physically correct.

Table 3: Agentic evaluation under the physics-constrained reward. Values are means across 90 rollouts per policy; rates are computed from archived governance logs.

Policy	Audit rate	Revision rate	Action modified	Mean action delta
PPO	0.344	0.943	0.987	0.164
SAC	0.342	0.957	0.524	0.059
DQN	0.372	0.955	0.518	0.084

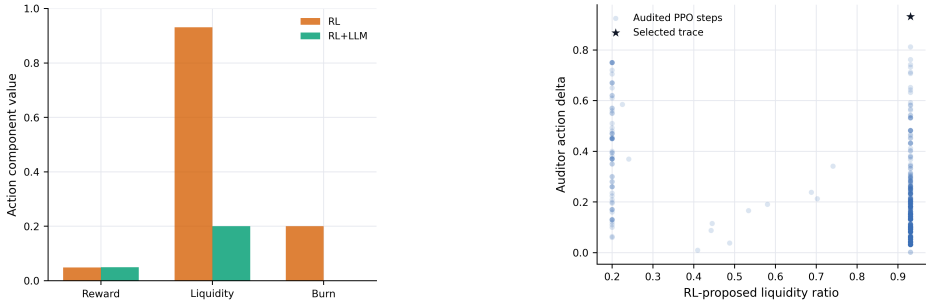


(a) Auditability rates. (b) Action-jitter delta. (c) Auditor action delta.

Figure 7: Agentic governance diagnostics for the LLM Planner/Auditor layer. The original auditability, paired episode-level delta, and step-level Auditor action-delta images are displayed with matched subfigure heights.

Credible generation records matter directly to prosumers and the energy communities in which they trade. Market operators must also understand how automated allocation changes liquidity and settlement, while regulators need records that connect a contested decision to the action that was proposed and executed. SolarChain-Eval offers a common evidence schema for inspecting these failure modes before deployment. The present experiments do not establish population-level distributional effects, climate outcomes, or real-world grid reliability. Connections to SDGs 7, 9, 10, and 13 are therefore prospective rather than empirical findings of this study.

Several limitations bound these conclusions. The evaluation uses one monthly five-city benchmark instance, and its simulator simplifies market clearing and token-price dynamics. The LLM endpoint is not assumed to be bitwise deterministic, although structured outputs and accepted logs are retained. Seed 20260511 was produced immediately before a progress-display-only code change; the change is recorded and does not alter the scientific components. The study does not cover other operating regimes or grid constraints, nor does it vary reward weights and adversarial frequency or calibrate against real-world market data. Future work should compare constrained-RL baselines and test the Auditor against safe-action holdouts so that false-positive and false-negative rates can be measured. Broader weather and market traces, followed by validation with energy-market operators, are also needed. These are research extensions, not evidence that can be inferred from the present artifacts.



(a) Selected action revision. (b) Audited PPO context.

Figure 8: Representative audit trace from the archived agentic logs. These original images show the action correction and the market context that activated the audit.

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A Core Reproducibility Record

This appendix records only the evidence boundary needed to interpret and verify the reported results. All numerical summaries derive from the three completed seed archives; no policy was retrained and no experiment was rerun for this revision.

Table 4: Core artifact and evaluation record. Counts refer to archived observations used in the paper.

Component	Preserved evidence
Market instance	Five cities, 50 nodes, 720 hourly states, 36,000 generation rows, 1,800 FDIA-marked rows, and 1,185 P2P trades.
Training	PPO, SAC, and DQN checkpoints after 300,000 timesteps for seeds 20260511, 20260512, and 20260513; Static, Random, and Myopic are non-learning controls.
Evaluation unit	Thirty 24-hour episodes per seed, yielding 90 rollout-level observations per policy and setting. Tables report mean $\pm$ standard deviation.
Settings	Physics-constrained RL, no-physics RL, physics-constrained RL+LLM, and no-physics RL+LLM. The Planner/Auditor is evaluation-only.
Pairing	Ablation and governance deltas match seed, policy, and episode before subtraction.
Agentic evidence	2,160 hourly governance records per learned policy and agentic setting, including triggers, proposed and final actions, decisions, deltas, and rationales.
Aggregate boundary	1,620 rollout-metric rows and 12,960 governance-log rows. Configurations, checkpoints, logs, and figure scripts are retained in the accompanying anonymous artifact package.

The physics oracle computes suspicious supply from PV upper-bound exceedance or FDIA labels, while the deterministic sanitizer enforces component bounds and the global allocation budget. The no-physics ablation removes only the physics penalty from optimization and continues to log physical risk. Consequently, the appendix supports reproducibility of the descriptive and paired comparisons but does not extend the claims to causal governance effects, unseen operating regimes, or real-world grid performance.